DSP · L06

DFT and circular convolution

Answers and worked solutions · Day 09 · Sun 1 Nov · pack D5

5. Practice

From your course files

T5 Q1 (tutorial).

> *Hint: Playbook D. Row 0 of the circulant is* $\{x_2(0), x_2(3), x_2(2), x_2(1)\} = \{4, 2, 2, 3\}$.
>
> $y(n) = x_1(n) \circledast_4 x_2(n) = \sum_{m=0}^{3} x_1(m) x_2((n-m))_4$, $n = 0, 1, 2, 3$:
>
> - $y(0) = x_1(0)x_2(0) + x_1(1)x_2(3) + x_1(2)x_2(2) + x_1(3)x_2(1) = 4 + 4 + 6 + 3 =$
> - $y(1) = x_1(0)x_2(1) + x_1(1)x_2(0) + x_1(2)x_2(3) + x_1(3)x_2(2) = 3 + 8 + 6 + 2 =$
> - $y(2) = x_1(0)x_2(2) + x_1(1)x_2(1) + x_1(2)x_2(0) + x_1(3)x_2(3) = 2 + 6 + 12 + 2 =$
> - $y(3) = x_1(0)x_2(3) + x_1(1)x_2(2) + x_1(2)x_2(1) + x_1(3)x_2(0) = 2 + 4 + 9 + 4 =$
>
> $$\mathbf{y(n) = \{\underset{\uparrow}{17}, 19, 22, 19\}}$$

T5 Q2 (tutorial).

> *Hint: Playbook B then D step 4. Do* $X(0), X(1), X(2)$*; get* $X(3) = X^*(1)$.
>
> Recall $x_1(n) \circledast_4 x_2(n) \overset{\text{DFT}_4}{\longleftrightarrow} X_1(k)X_2(k)$.
>
> - $X_1(k) = \sum_{n=0}^{3} x_1(n) e^{-j2\pi kn/4} = \{7,\ -2 - j,\ 1,\ -2 + j\}$
> - $X_2(k) = \{11,\ 2 - j,\ 1,\ 2 + j\}$
> - $X_1(k)X_2(k) = \{77, -5, 1, -5\}$ (e.g. $(-2 - j)(2 - j) = -4 + 2j - 2j + j^2 = -5$)
> - $y(n) = \frac{1}{4}\sum_{k=0}^{3} Y(k) e^{j2\pi kn/4}$: $y(0) = \frac{1}{4}(77 - 5 + 1 - 5) = 17$, $y(1) = \frac{1}{4}(77 - 5j - 1 + 5j) = 19$, $y(2) = \frac{1}{4}(77 + 5 + 1 + 5) = 22$, $y(3) = 19$.
>
> $$\mathbf{y(n) = \{\underset{\uparrow}{17}, 19, 22, 19\}}$$
>
> the same as Q1.

T5 Q3 (tutorial).

> $x(n) = \{\underset{\uparrow}{1}, 1, 1, 1, 0, 0, 0, 0\}$, so
>
> $$X(k) = \sum_{n=0}^{7} x(n) e^{-j2\pi kn/8} = 1 + e^{-j\pi k/4} + e^{-j\pi k/2} + e^{-j3\pi k/4}$$

$$\mathbf{X(k)} = \{\mathbf{4},\ \mathbf{1-j(1+\sqrt{2})},\ \mathbf{0},\ \mathbf{1+j(1-\sqrt{2})},\ \mathbf{0},\ \mathbf{1-j(1-\sqrt{2})},\ \mathbf{0},\ \mathbf{1+j(1+\sqrt{2})}\}$$

$x_1(n) = x((n+3))_8 = x((n-5))_8$, so the circular time-shift property $x((n-l))_N \leftrightarrow X(k)e^{-j2\pi kl/N}$ gives $\mathbf{X_1(k) = X(k)e^{j3\pi k/4}}$. Likewise $x_2(n) = x((n-2))_8$, so $\mathbf{X_2(k) = X(k)e^{-j\pi k/2}}$.

T5 Q6 (tutorial).

(a) $X(\omega) = \sum_{n=-2}^{3} x(n)e^{-j\omega n} = \mathbf{3 + 4\cos\omega + 2\cos(2\omega)}$ (the pairs $2e^{\pm j\omega}$ and $1 \cdot e^{\pm j2\omega}$ combine by Euler; $x(3) = 0$).

(b) $V(k) = \sum_{n=0}^{5} v(n)e^{-j2\pi kn/6} = 3 + 4\cos\dfrac{\pi k}{3} + 2\cos\dfrac{2\pi k}{3} = \{\mathbf{9, 4, 0, 1, 0, 4}\}$.

(c) $\mathbf{V(k) = X(\omega)|_{\omega=2\pi k/6}}$: $V(k)$ samples $X(\omega)$ at $N = 6$ equally spaced frequencies. In the time domain, sampling $X(\omega)$ corresponds to the periodic extension $x_p(n) = \sum_l x(n - lN)$ with $N = 6$, i.e. $\{\ldots, 1, 2, \underset{\uparrow}{3}, 2, 1, 0, 1, 2, 3, \ldots\}$, and one period of it starting at $n = 0$ is $v(n) = \{\underset{\uparrow}{3}, 2, 1, 0, 1, 2\}$.

T5 Q4 (tutorial).

The inverse system is $H_I(z) = 1 - 0.5z^{-1}$, with $h_I(n) = \{\underset{\uparrow}{1}, -0.5\}$, so $x(n) = h_I(n) * y(n)$.

1. Zero-pad h_I to length 64: $\bar{h}_I(n) = \{h_I(n), 0, \ldots, 0\}$ (62 zeros). $y(n)$ is already length 64.
2. Compute $\bar{H}_I(k)$ and $Y(k)$ (64-point DFTs), then $\bar{h}_I(n) \circledast_{64} y(n) = \mathrm{IDFT}_{64}\{\bar{H}_I(k)Y(k)\}$.
3. For $n = 1, \ldots, 63$: $\bar{h}_I \circledast_{64} y = h_I(0)y(n) + h_I(1)y(n-1)$, which is the linear convolution, so it equals $x(n)$.
4. For $n = 0$ the circular result is $h_I(0)y(0) + h_I(1)y(63) = y(0) - 0.5y(63)$, but the linear one is $h_I(0)y(0)$ (with $y(n) = 0$ for $n < 0$). Correct it: $\mathbf{x(0) = \left[\bar{h}_I \circledast_{64} y\right]_{n=0} + 0.5\,y(63)}$.

New questions

Q1 (easy) [3 marks].

$X(0) = 1 + 0 + 2 + 1 = 4$ $X(1) = 1 + 0(-j) + 2(-1) + 1(j) = -1 + j$
$X(2) = 1 - 0 + 2 - 1 = 2$ $X(3) = 1 + 0(j) + 2(-1) + 1(-j) = -1 - j$

$\mathbf{X(k) = \{4,\ -1+j,\ 2,\ -1-j\}}$. $X(3) = -1-j = (-1+j)^* = X^*(1)$, as expected for real x. Check: $\frac{1}{4}(4-1+j+2-1-j) = 1 = x(0)$.

Q2 [4 marks].

a. Rows of 4: $\{1,2,3,4\}$, $\{5,6,0,0\}$. Column sums: $\mathbf{\tilde{x}(n) = \{6,8,3,4\}}$.

b. $\tilde{X}(0) = 6+8+3+4 = 21$ and $X(0) = 1+2+\cdots+6 = 21$. Equal, because the samples of $X(\omega)$ are exact even with aliasing; only the time sequence is aliased.

c. $N \geq L = 6$, so $\mathbf{N = 6}$.

Q3 [5 marks].

a. Pad x_2 to $\{1,1,1,0\}$. Circulant (column 0 = x_2):

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+0+0+1 \\ 1+2+0+1 \\ 1+2+0+0 \\ 0+2+0+1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 3 \\ 3 \end{bmatrix}$$

$\mathbf{y_4 = \{2,4,3,3\}}$.

b. Linear: $y = \{1,3,3,3,1,1\}$ (length $4+3-1=6$). Fold $y(4), y(5)$ onto $n = 0, 1$: $\{1+1, 3+1, 3, 3\} = \{2,4,3,3\}$. Matches (a). Check: $\sum y = 12 = 4 \times 3$.

Q4 [6 marks, chained].

a. $L + M - 1 = 3 + 2 - 1 = \mathbf{4}$.

b. $x = \{1,2,1,0\}$: $X(0) = 4$, $X(1) = 1-2j-1 = -2j$, $X(2) = 1-2+1 = 0$, $X(3) = 2j$. So $X = \{4, -2j, 0, 2j\}$. $h = \{1,-1,0,0\}$: $H(k) = 1-(-j)^k$: $H = \{0,\ 1+j,\ 2,\ 1-j\}$.

c. $Y = \{0,\ -2j(1+j),\ 0,\ 2j(1-j)\} = \{0,\ 2-2j,\ 0,\ 2+2j\}$.
$y(n) = \frac{1}{4}\sum_k Y(k) j^{kn}$: $y(0) = \frac{1}{4}(4) = 1$;
$y(1) = \frac{1}{4}[(2-2j)j + (2+2j)(-j)] = \frac{1}{4}(2j+2-2j+2) = 1$;
$y(2) = \frac{1}{4}[-(2-2j) - (2+2j)] = -1$;
$y(3) = \frac{1}{4}[(2-2j)(-j) + (2+2j)j] = \frac{1}{4}(-2j-2+2j-2) = -1$.
$\mathbf{y(n) = \{1,1,-1,-1\}}$. Direct: $\{1,2,1\} * \{1,-1\} = \{1,1,-1,-1\}$. Agrees.

d. $y(n)$ has length $4 \leq N$, so its 4-point DFT is exactly its DTFT sampled at $\omega_k = 2\pi k/4 = \pi k/2$ (no time aliasing). With $N = 3$, $y(3) = -1$ wraps onto $n = 0$: $\mathbf{\{0,1,-1\}}$.

Q5 [6 marks].

a. $X_a(k) = X(k)W_4^k = X(k)(-j)^k$: $k=0$: 10; $k=1$: $(-2+2j)(-j) = 2+2j$; $k=2$: $(-2)(-1) = 2$; $k=3$: $(-2-2j)(j) = 2-2j$.
$\mathbf{X_a = \{10,\ 2+2j,\ 2,\ 2-2j\}}$. (Check: $x_a = \{4,1,2,3\}$, $X_a(2) = 4-1+2-3 = 2$.)

b. $x_b = \{1,4,3,2\}$, $X_b(k) = X((-k))_4 = X(4-k)$:
$\mathbf{X_b = \{10,\ -2-2j,\ -2,\ -2+2j\}}$. For real x this is also $X^*(k)$.

c. $(-1)^n = e^{j2\pi(2)n/4}$, so $X_c(k) = X((k-2))_4$:
$\mathbf{X_c = \{-2,\ -2-2j,\ 10,\ -2+2j\}}$.

d. $\frac{1}{4}(100+8+4+8) = \mathbf{30} = 1+4+9+16$.

Q6 (hard) [6 marks].

a. $N = 4 \geq$ length of y. $Y(k)$: $Y(0) = 15$; $Y(1) = 2-5j-6+2j = -4-3j$; $Y(2) = 2-5+6-2 = 1$; $Y(3) = -4+3j$. $H(k) = 2+(-j)^k$:
$H = \{3,\ 2-j,\ 1,\ 2+j\}$. All nonzero, so division is allowed. $X(k) = Y/H$:
$X(0) = 5$; $X(1) = \frac{-4-3j}{2-j} = \frac{(-4-3j)(2+j)}{5} = \frac{-5-10j}{5} = -1-2j$; $X(2) = 1$;
$X(3) = -1+2j$. IDFT: $x(0) = \frac{1}{4}(5-1-2j+1-1+2j) = 1$;
$x(1) = \frac{1}{4}[5+(-1-2j)j-1+(-1+2j)(-j)] = \frac{1}{4}(5+2-j-1+2+j) = 2$
; $x(2) = \frac{1}{4}[5+1+2j+1+1-2j] = 2$;
$x(3) = \frac{1}{4}[5+(-1-2j)(-j)-1+(-1+2j)j] = \frac{1}{4}(5-2+j-1-2-j) = 0$
. $\mathbf{x(n) = \{1,2,2\}}$. Check: $\{1,2,2\} * \{2,1\} = \{2,5,6,2\}$.

b. $h = \{1,1,0,0\}$ gives $H(k) = 1+(-j)^k = \{2, 1-j, 0, 1+j\}$. $H(2) = 0$, so $X(2) = Y(2)/0$ is undefined. The filter removes the $\omega = \pi$ component, so that part of x cannot be recovered at this N.

6. Quick recall check

Question

1. 2.
2. $[1,\ -j,\ -1,\ j]$.
3. $10+5-1 = 14$ (16 if a radix-2 FFT is required).
4. $X(5) = X^*(3) = 1-2j$.
5. $N^2 = 65\,536$ vs $\frac{N}{2}\log_2 N = 1024$ (64 times fewer).